Elisabetta Fedele

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Education

ETH Zürich — **Zürich, Switzerland**
PhD in Computer Science — *Sep 2024 – present*
- Co-advised by Prof. Marc Pollefeys and Prof. Bernhard Schölkopf
- Supported by the ETH AI Center Fellowship *(2% acceptance rate)*

ETH Zürich — **Zürich, Switzerland**
Master in Computer Science — *Sep 2021 – July 2024*
- Major in **Machine Intelligence** and Minor in **Theoretical Computer Science**.
- **Courses**: Algorithmic Game Theory, Algorithms Lab, Randomized Algorithms and Probabilistic Methods, Graph Theory, Computational Intelligence Lab, Generative Visual Models, Advanced Machine Learning, Machine Perception, Computer Graphics, Computer Vision, NLP.

Politecnico di Milano — **Milan, Italy**
Bachelor in Computer Systems Engineering — *Sep. 2018 – Jun. 2021*
- Final grade: 110/110 *cum laude*.

Experience

Meta Reality Labs — **Menlo Park, USA**
Research Intern — *Aug. 2026 – present*
- Advised by Lea Wilken and Jeannette Bohg.
- Working on Agentic Robot Learning.

Stanford University — **Stanford, USA**
Research Intern — *Mar. 2025 – Aug. 2025*
- Advised by Leonidas Guibas and Francis Engelmann.
- Worked on primitive-based representations of 3D scenes and on spatial conditioning for 3D generation.
- Resulted in first-author publications at **ICCV 2025** (**Oral**) and **ICLR 2026**.

ETH Zürich — **Zurich, Switzerland**
Master Thesis Student — *Feb. 2024 – July 2024*
- Advised by Marc Pollefeys and Francis Engelmann.
- Worked on primitive-based representations of 3D scenes.

EPFL — **Lausanne, Switzerland**
Research Intern — *Jul. 2023 – Sep. 2023*
- Advised by Alessandro Chiesa and Giacomo Fenzi.
- We found a new family of algorithms that greatly improves the asymptotic and practical performance of the sumcheck protocol. Recently, these algorithms were proven to be optimal.
- Resulted in a publication at *Theory of Cryptography Conference* (**TCC**) **2025**.

Google Research — **Zurich, Switzerland**
Semester Project Student — *Mar. 2023 – Jun. 2023*
- Advised by Federico Tombari, Marc Pollefeys, and Francis Engelmann.
- We leveraged the capabilities of CLIP to expand the task of 3D instance segmentation to an Open-Vocabulary setting.
- Resulted in a publication at **NeurIPS 2023**.

Politecnico di Milano — **Milan, Italy**
Research Assistant — *Oct. 2020 – Mar. 2021*
- Advised by Stefano Ceri.
- We processed and analyzed data on SARS-CoV2 viral sequence mutations at the nucleotide and amino acid levels.

Publications and Preprints

- **Elisabetta Fedele**, René Zurbrügg, Lior Yariv, Or Litany, Bernhard Schölkopf, Leonidas Guibas, Marc Pollefeys, Francis Engelmann. "PixelDec: Generative Scene Decomposition with Superquadrics from a Single Image". *Under review.*
- Neil De La Fuente, Joan Lafuente Baeza, Mukhammadali Sayfiddinov, Felicia Scharitzer, Marc Pollefeys, Ata Celen, Sayan Deb Sarkar*, **Elisabetta Fedele***. "SpaceFlow: Locally Controllable 3D Generation". *Under review.*
- Gabriel Tavernini*, **Elisabetta Fedele***, Tiago Novello, Leonidas Guibas, Marc Pollefeys, Francis Engelmann. "SuperFlex: Deformable Superquadrics for Point Cloud Decomposition". *European Conference on Computer Vision (**ECCV**) 2026.*
- **Elisabetta Fedele***, Francis Engelmann*, Ian Huang, Or Litany, Marc Pollefeys, Leonidas Guibas. "SpaceControl: Introducing Test-Time Spatial Control to 3D Generative Modeling". *International Conference on Learning Representations (**ICLR**) 2026.*
- **Elisabetta Fedele**, Boyang Sun, Leonidas Guibas, Marc Pollefeys, Francis Engelmann. "SuperDec: 3D Scene Decomposition with Superquadric Primitives". *International Computer Vision Conference (**ICCV**) 2025* (**Oral**, top 2.4%).
- Dimitri Von Rütte*, **Elisabetta Fedele***, Jonathan Thomm*, Lukas Wolf*. "FABRIC: Personalizing Diffusion Models with Iterative Feedback". *European Conference on Computer Vision Workshops (**ECCVW**) 2024.*
- Ayca Takmaz*, **Elisabetta Fedele***, Robert W. Sumner, Marc Pollefeys, Federico Tombari, Francis Engelmann. "OpenMask3D: Open-Vocabulary 3D Instance Segmentation". *Advances in Neural Information Processing Systems (**NeurIPS**) 2023.*

Academic Activities and Outreach

- **Co-organizer of OpenSUN3D**, *ICCV'23'25, ECCV'24, CVPR'24'25'26* 2023-present
  I am among the organizers of OpenSUN3D, a series of Workshops on Open-Vocabulary 3D Scene Understanding, held twice a year in conjunction with the major computer vision conferences.
- **Co-organizer of ZurichCV Meetups**, *Zurich* 2024-present
  I am among the organizers of the ZurichCV Meetups, a monthly meetup which gathers the Computer Vision community of Zurich both from academia and industry.
- **Conference reviewer** 2024-present
  Reviewed papers for workshops and main conferences at SIGGRAPH, SIGGRAPHASIA, ICCV, ECCV, CVPR, 3DV, ICLR, ICRA.

Honors & Awards

- **SwissAI Grant for Small Projects**, *SwissAI*, 50k GPU hours 2025
- **NVIDIA Academic grant**, *NVIDIA*, USD 100k cloud credit, 12k hardware 2025
- **SNSF Mobility Grant in Projects**, *Swiss National Science Foundation (SNSF)*, CHF 20k 2025
- **ETH AI Center Doctoral Fellowship**, *ETH AI Center*, 2% acceptance rate 2025
- **ICVSS Best Presentation Award**, *International Computer Vision Summer School (ICVSS)* 2024
- **NeurIPS 2023 Scholar Award**, *NeurIPS* 2023
- **Summer@EPFL Fellowship**, *EPFL*, 1.5% acceptance rate 2023
- **Scholarship for Sardinian students**, *Regione Sardegna* 2022-2023
- **Merit-Based Scholarship**, *Politecnico di Milano* 2020-2021
- **Scholarship for Sports Merits**, *Politecnico di Milano* 2019
- **Best Freshmen Prize**, *Politecnico di Milano* 2019

Extracurricular activities

Sailing Instructor Diploma - Level I, *Italian Sailing Federation* 2022 - present

Professional Optimist and 420 sailor, Cagliari/Mandello del Lario 2012–2019

- Member of the **Italian Sailing Team** from 2012 to 2019.
- Competed in 6 **World Championships** and in 4 **European Championships**.
- Awarded **Best Italian Female Sailor** by *Tutto Vela* for the year 2014.

* authors contributed equally